Python Basics Notes

What is Python?

Python is a high-level, interpreted programming language that is easy to learn and widely used for web development, data science, artificial intelligence, automation, and scripting.

Variables

Variables store data values. Example: x = 10, name = 'Alice'.

Data Types

Common data types include int, float, str, bool, list, tuple, dictionary, and set.

Operators

Python supports arithmetic (+,-,*,/), comparison (==, !=, >, <), logical (and, or, not), and assignment operators.

Conditional Statements

Use if, elif, and else to execute code based on conditions.

Loops

A for loop repeats over a sequence. A while loop repeats while a condition is true.

Functions

Functions are reusable blocks of code created using the def keyword. They can take parameters and return values.

Lists

Lists are ordered, mutable collections. Example: fruits = ['apple','banana','mango'].

Dictionaries

Dictionaries store key-value pairs. Example: student = {'name':'Dhanush','age':21}.

Conclusion

Python is beginner-friendly and powerful, making it an excellent first programming language.